# Operational support systems for an intelligent network

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*Introducing services based upon intelligent networks (IN) has substantial implications for management systems that support those services and networks. This paper summarises what operational support systems (OSSs) need to do, and then goes on to examine the challenges for OSSs within an IN context. It then gives some possible ways forward to meet those challenges.*

## 1. Introduction

Most of the papers in this issue of the Journal concentrate upon the features that the network systems provide and how the network provides them. However, in order to offer a usable service to customers, it is also necessary to manage the network. This is the job of operational support systems (OSSs).

In a functional (or logical) architecture the functions provided by OSSs are broken up into a number of blocks, each with a limited range of closely associated functions, particularly with respect to the data upon which those functions act. Until recently, the high-level functional architecture accepted across the business was that produced originally by CNA [1] and subsequently modified somewhat by the BT Business Model [2]. This splits BT's business functions into four key blocks, as identified in Fig 1.

For the purposes of operational support systems, the key functions from that architecture are those of network and service management. Business management is mainly concerned with how the business is run and is therefore to do with business strategy and information. Resource management is about business resources and is therefore to do with work management, inventory, etc.

### *1.1 Service management*

Service management is principally concerned with how customers see the service being offered. In a reduced form, it can be thought of as how service is provided, maintained, monitored and billed (PMMB being a useful abbreviation to remember). The full set of functions is illustrated in Fig 2 [3].

The architecture identifies each of the functions that need to be managed in any service. While it may be obvious to most people that billing and ordering should be a key consideration when introducing a new service, issues such as reporting the performance of the service (service monitoring), handling problems (service maintenance) and producing sales material (marketing operations) are more easily forgotten.

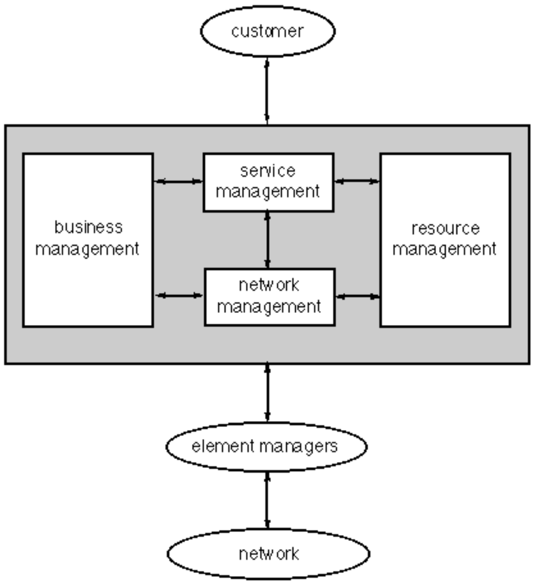


Fig 1 BT's logical architecure.

### *1.2 Network management*

If service management is about how to manage the service, and therefore the interface to the customer, network management is concerned with the interface to the network.

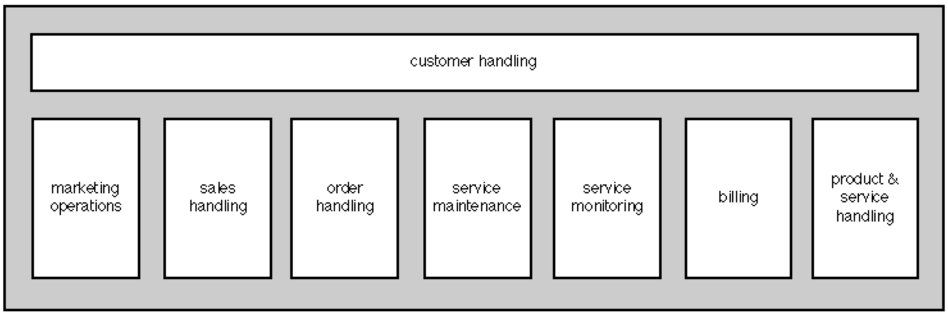


Fig 2 BT's service management logical architecture.

It is principally concerned with issues such as network faults, configuration of network equipment (such as switches, multiplexers), network planning, etc.

In the same way that each new service needs all the functions of service management to be considered, each network needs all the functions of network management to be considered. It should be noted that the two need not be the same — to provide one service can require many networks, and one network can offer many services. For example a future IN network would be expected to offer a huge multiplicity of services.

### *1.3 Why architectures?*

A logical architecture shows what must be done in order to manage services and networks. This clear knowledge of the range of management functions provides a baseline to assess the impact of new service types and network technology. An architecture will be a key feature of new OSS designs to meet the challenge of services based on intelligent networks.

## 2. The challenge of IN

The intelligent network concepts originally developed by Bellcore [4] successfully separated service and feature control from the physical aspects of the network. Service logic embedded in attached processors (e.g. SCPs) permits changing the flow of a basic telephone call to provide enhanced communications services.

However, this separation increases the complexity of management. Connectivity is established through changes to relationships between elements of hardware, software and the links between them, and this involves rules, conventions and processes for establishment and termination of such relationships. Failure modes are also more complex, points of service failure are no longer fixed and are not easy to test in the operational network.

For example consider a network-based automated call-distribution feature. This routes calls to a number of sites based on time of day, call load and country of originating call. How does the telco test the service in the event of reported faults? Which of a number of networks are at fault? Is it a physical or a logical fault? Is it the service logic routeing the calls to the network database? Knowledge of previous call load is necessary to determine the correct routeing of any particular call. Will it be acceptable for a BT customer who reports a fault on one of its sites, where the access service is provided by telco X, to be told by BT that BT's network is alright and that telco X should be contacted?

Thus IN services will require management of new physical network elements (the attached processors) and complex configuration of service logic and databases. End-to-end management of communications services across multiple networks will also be required if truly seamless services are to be offered.

However, locating and fixing faults in an intelligent network is not the only new demand on the management systems. For example, the following are key features of IN-based services, all of which create new problems for management systems:

- rapid introduction of new services,
- open-ended range of services,
- volatility of customer and service data,
- increased customer control,
- enhanced signalling systems,
- third party service provision,
- distributed customer and network data,
- new network elements.

### 2.1 Rapid service introduction

A key driver for the introduction of intelligent networks is their ability to allow operators to design and launch new services quickly, in order to meet the increasing pace of the changes in the needs of the market-place. Within an IN context, this process is referred to as service creation and more detail of the process can be found in Turner [5].

However, simply designing and launching the service on the network leads to a service for which problems cannot be managed, orders cannot be taken and bills cannot be produced. Clearly, the network enhancement must be accompanied by enhancements to management systems in the same time-scales. Historically, management systems have taken in the order of a year to enhance, whereas intelligent networks are looking at new service launch times in the order of days. Therefore the way services are currently managed will need to be substantially changed if these two differing time-scales are to be brought together [6].

If rapid service introduction is to succeed, business processes will also need to change. At present, the processes presuppose a long product launch cycle — much work will be needed in this area to improve response times.

### 2.2 Open-ended range of services

Currently, the range of services offered on BT's network platform is well bounded and understood. However, intelligent networks allow the possibility of a vast number of new and innovative services being created quickly. This can lead to an uncontrolled range of services being available. This causes a number of problems, principally concerned with feature interaction, but increasingly in the control of the range of services offered.

Furthermore, next generation services will not generally be provided individually as low-level features. They will be combined increasingly in 'packages'. These packages can be a mixture of some customer premises equipment and one or more sophisticated network services.

For example, consider chargecard access to a global corporate virtual private network. To be effective, management must span the entire system across local, national and international boundaries and across the numerous vendor systems likely to be deployed. This will present new challenges to the designers of management systems. They will need the flexibility to manage services in a number of different combinations and still provide compatible operations such as fault reporting and correlation.

### 2.3 Volatility of customer and service data

IN services are characterised by the rapidly changing nature of the customer and service data [7]. For example, a number translation service is designed so that the dial plan can be changed rapidly, say, in the event of a sudden surge of calls. Managing and tracking this changed data provides special problems. Clearly, volatile data is held on the service control point (SCP), but it is likely that it will also need to be held on the management systems. For example, fault finding, billing, and customer reporting will all need access to that data. This means that data management strategies are required which allow volatile service data to be maintained, whilst managing different versions of replicated data.

### 2.4 Increased customer control

The complexity of IN services, and the increasing business dependence upon them, means that customers will increasingly want to control their own network services, e.g. to add new extensions to a centrex service or to increase the bandwidth of a data service. This adds substantial complexity to the management systems that the telco will need to provide. Enhanced security will be a key need, both to prevent the network being brought down by uncontrolled or unauthorised changes, and to prevent customers meddling with each others' service configuration. The interface to the customer will also need to be much more user-friendly than is customary with management systems, in order to avoid huge training costs.

### 2.5 Enhanced signalling systems

The IN requirement for complex signalling [8] based around CCITT signalling system No 7 also has implications for managing the signalling network. Since traffic routeing becomes dependent upon signalling, the characteristics of the signalling network changes. This means that the signalling network itself will require new and quicker acting management methods, and creates a need for separate management of the signalling network. The signalling load and integrity requirements also grow.

### 2.6 Third party service provision

The window of opportunity to launch a service and recoup the investment is shortening. An increasing number of applications may be only appropriate for niche markets. These may not be economical if a telco has to bear the full cost of service development and deployment. Such services may be provided in collaboration with third party service developers who can develop these services and sell them to network operators. This could enable telcos to meet their customer demands more quickly and at a lower overall cost. In addition, non-telecommunications services (e.g. interactive games) may be offered through the public network of the future.

The introduction of intelligent networks is the first use of an architecture in which a third party may gain direct access into the public telecommunications system. This represents a significant evolution from conventional switching architectures and will, in turn, require a corresponding change in thinking about securing the operation of such networks.

The extent of interconnection between telcos' intelligent networks remains to be determined, with regulatory bodies such as the DTI [9], European Commission and FCC in the USA certain to influence the outcome. Other uncertainties persist in areas such as future EC directives on the rights of individuals to privacy.

A telco could suffer substantial losses if the limit of open access to third parties is not clearly defined within a secure management framework. As well as the effects of fraud and intentional disruption of the network, losses could result from accidental interference with the signalling network or perhaps from unforeseen service interactions.

Both ETSI and the ITU have working parties addressing this issue, and there is work in progress within the Eurescom programme. However, the results of these initiatives is not expected for several years. In the meantime, telcos will implement bespoke countermeasures based on an assessment of service and network asset evaluations and the vulnerability of the physical and software infrastructure to recognised threats and risks.

### *2.7 Distributed customer and network data*

With a non-IN network, details of a customer's routeing and network routeing data are localised to a single switch. Even if more advanced services than simple call delivery are employed (e.g. call diversion), only the data in the local switch needs changing. However, IN changes all that. If a customer uses a personal mobility service to change his delivery location from London to Sydney, then a customer calling from Sydney should not be routed through to London before being diverted back. This implies that the routeing plan has to be accessible to all locations on the network. This, in turn, means either a distributed database or a vast, single database with access from anywhere in the world. While this is currently possible on the scale needed to support VPNs for large businesses, neither would be currently possible on the scale which would be needed to provide service to a mass market. The former solution is clearly more scalable, but taxes distributed computing technology to its limits; the latter solution would require a huge central processor and has communications requirements that would be difficult to meet.

### *2.8 New network elements*

Intelligent networks bring with them new classes of network elements, and new issues of managing them. The key new system is the service control point (SCP), which is a large computer, sourced from IT vendors and usually based upon fault-tolerant technology. It is usually optimised for rapid database access. The difference between SCPs and standard stored program control (SPC) exchanges (e.g. System X) is in the degree of specialisation. The SPC control computer is dedicated to the job of managing the switch. However, an SCP is a commodity component specialised for a particular job. Therefore managing the SCP as a part of a network starts to introduce computer systems management problems to what had been previously a transport network.

## 3. Other market drivers

The challenge for OSS design and implementation is not just to manage IN based networks — the market is also changing in other areas. OSSs must combine the capability to manage IN services whilst providing the following capability.

### *3.1 Outsourcing*

Those corporations who wish to continue building and operating their own private networks increasingly will demand manageable network components and network services. Others will want a smooth and safe migration to a point where a telco, or outsourcing company, is running all or part of the customer's network and performing extensions, upgrades, changes, etc, for a predictable fee. Some customers, for example, may want to retain operations of the North American segment of their virtual private network but use an outside source to provide the European nodes. The split may be based on other criteria, e.g. the time of day. The management of this telecommunications package is likely to be a shared activity between network operator and the customer — in other words a co-operative activity between the two. This sharing will need to be on a tailored basis. However, in general, most customers will want to concentrate on their business and service management aspects whilst the telco handles the operational issues of managing the network on a day-to-day basis within agreed service levels.

Finally a customer may want to sub-contract the entire provision and operation of its information network to a managed services operator. The customer will want to dictate different service level agreements with the telco or operator for various network applications.

### *3.2 The market*

Society is becoming increasingly dependent upon telecommunications services. A study conducted in the USA by the University of Minnesota [10] estimates that the loss of voice and data transmission capability could completely shut down a bank in two days and a factory in five. Just one hour without telecommunications services could cost an insurance company $20,000, an airline $2.5 million and an investment bank $6 million. Hence very high service levels will be required for high-value applications and much lower service (and hence lower cost) levels for less vital functions with the ability to change these priorities based on business demand.

The customer will want to dictate the locations to be served, the capacity between sites and service features to be available at those sites and may want to vary the price/performance mix of the telecommunications services by time of day/year or geography. They will demand management information to improve their decision-making processes.

This type of flexible service needs an electronic form of service level management between the customer and the telco. In order to implement these requests the operator (and the customer if continuing to operate part of the network directly) needs an integrated structure between this service view of the network and the network itself so that control can be exercised with minimum human intervention. This leads to a layered architecture for the management of customer network services and the complex information which flows between the telco and the customer.

### *3.3 Correspondent working*

Currently liberalisation and deregulation of telecommunications services has not advanced to the point where a single telco is allowed to own or control all the elements required to provide access to customers worldwide. So a telco's global network is currently developed in conjunction with other operators throughout the world (known as correspondents). Within the UK, customers will increasingly use multiple network operators for their communications services. Hence the international call model, whereby BT is only responsible for a proportion of the call, will become more common within the UK. Interworking and collaboration with other network operators will have to be established. For example, the traffic management problems of a telethon or a phone-in which generates very large numbers of calls directed at a single answering point, will not be completely under BT's control. Without the visibility and control of the network provided by 'collaborative' network traffic management these events could lead to significant network failures and outages.

The challenges this brings will be similar to those posed by the move to personal mobility services. The ability of a customer to move between the fixed, mobile analogue, GSM (global system for mobility) and other more localised mobile networks will mean that the operational issues of such networks must be addressed. Not only must the networks physically interwork, the operational support systems of such networks must be able to interwork seamlessly, so that service provision, maintenance and billing appear unified for the customer, even though the networks themselves may be owned by different operators.

## 4. Enabling technologies

The challenge for OSSs in the future can be summed up as the need to be global, rapidly reconfigurable and providing user-friendly interfaces to customers. At present, there is no answer to that challenge. However, a number of enabling technologies have been identified and are being worked on.

### *4.1 Service packaging*

Services built on an IN will increasingly rely upon sets of common building blocks, which can be rapidly combined to provide a new service. For example, by combining a 'time of day' package with an 'origin of call' package, a call-answering bureau service can be built, with calls being routed to the nearest office during the day, but sent to a central location off-peak. To allow management of this new service to be introduced as quickly as it can be configured on the network, the building blocks will need to have management data added to them. For example, pricing details for the package and performance reports that are associated with it would be included. This enhanced package is called a service feature. It is likely that more than one network building block could be built up into a service feature, to provide a higher level set of standard building blocks. The skill of the product manager and the engineering teams supporting him would then be to combine these standard building blocks in novel ways to meet customers' needs.

Once a set of common service features has been established, together with the management features that are associated with them, it will be possible to launch the management of those features on the management systems in the same way as the network features are launched on the IN.

### *4.2 Computing technologies*

The biggest trends in computing are currently towards putting significant computing power on users' desks, increasing use of client/server architectures for database access, the drive towards distributed computing environments and the cheap provision of high-bandwidth links.

This will lead to systems of increasing functionality and user-friendliness, but at a cost in terms of systems design complexity. Configuration management of large networks of personal computers is acknowledged to present significant problems, but where these systems are critical to managing the company's key services and networks, the need to solve these problems becomes essential. Migration strategies from current systems also need serious consideration.

### *4.3 Standards*

The next generation of BT's administration systems are being developed and procured within an architectural framework. It is in no customer's interests to be locked into one manufacturer's products and services where these cannot work with those of other suppliers. Today, the demand is for interoperable components with suppliers competing on quality and support, not proprietary architecture.

BT has responded to this demand from customers (and its own network) with the development of a co-operative approach to architectural specifications for all aspects of communications systems. This approach recognises the need for partnership, working alongside other vendors and customers to define truly interoperable systems that cope with today's multivendor, multi-location networking market-place. BT's Co-operative Networking Architecture takes a step beyond open systems by putting in place the specifications and agreements to ensure that real products and services will be able to work together and conform to national implementations of international standards. It is not static, but evolves with time to reflect changing business pressures and advances in technology.

As one of the largest network operators in the world, BT is investing heavily in systems built against Co-operative Networking Architecture for its own networks, and is now actively implementing these principles into its customer products and services. By harmonising the architecture and management capabilities of its products and services, BT will offer the customer a true end-to-end managed communications capability within its own portfolio and within multivendor networks.

This approach provides BT with a durable, flexible and economic base management platform capability and communications architecture, firmly based on accepted industry standards and value added components. This allows its management system designers to focus on developing those items in BT's applications portfolio which will reduce its cost base and differentiate its products and services.

### *4.4 Process re-engineering*

The consolidation of network administration into centralised operational units still requires extensive manual effort using systems that are not integrated. Future systems will be characterised by automatic flow-through whereby an event, be it a customer order or fault in the network, triggers off a set of actions in an integrated set of computer systems. These then automatically perform the relevant workstring to respond to that event. The processing of the workstring is such that the response has the minimal amount of human intervention necessary.

### *4.5 Advanced information processing*

As telecommunications networks get more complex, in order to support a much larger variety of services for its users, and associated management support systems grow in complexity, some of the functions of these complex network management systems can only be carried out by employing advanced information processing (AIP) [11 — 13] to make software systems more adaptable and intelligent. It is a combination of the fields of artificial intelligence and distributed information processing that provides many of the techniques that are necessary in developing AIP solutions to network management. Progress is such that these technologies can be expected to improve the type of software used in most operational support systems over the next five years.

## 5. The vision of the future

The vision of the future for the provision of IN-based services is already well established in many peoples' minds. It centres on the view of a product manager or marketing manager seeing a niche in the market for a new communications service one morning and then designing that service on a service creation environment on the same afternoon, prior to launching it on the network the next morning! However, how the management systems would cope with this is less well thought about. The vision that must accompany the one above is of service centres being able to take the first customer's order on the morning of the launch, of the first bill being sent out that evening, all based upon advertising and mailshots that were automatically despatched to likely customers for the service early on the morning of the launch. Clearly, network integrity will restrict the range of services that could be launched in this way, without significant engineering support. However, once that vision is possible, then the era of a rapid product launch ability could be said to have arrived.

## References


1 BT: 'Co-operative Network Architecture - Management', (1990).

2 'Service management systems: supporting the customer interface', British Telecommunications Eng J, 13, Part 1 (April 1994).

3 'Service management business model and logical architecture', internal BT document.

4 BELLCORE, SR-NPL-001555. AIN Release 1 baseline architecture (March 1990).

5 Turner G D: 'Service creation', BT Technol J, 13, No 2, pp 80—86 (April 1995).

6 Sinoyannis S: 'Service element-based operations requirements', IEEE Proc (1994).

7 Robruck R: 'The intelligent network — changing the face of telecommunications', IEEE Proc, 79 (January 1991).

8 Bale M C: 'Signalling in the intelligent network', BT Technol J, 13, No 2, pp 30—42 (April 1995).

9 Department of Trade and Industry: 'Consultation document on intelligent networks', and 'Intelligent networks — a response' (1992 and 1993).

10 AT&T 'Trends in telecommunications', 7, No 2, pp 23 (October 1991).

11 Helleur R J (Ed): 'Network management', BT Technol J, 9, No 3 (July 1991).

12 Flavin P G (Ed): 'Computer aided decision support', BT Technol J, 12, No 4 (October 1994).

13 Smith R, Azarmi N and Crabtree I B (Eds): 'Advanced information processing techniques for resource scheduling and planning', BT Technol J, 13, No 1 (January 1995).


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